

INDICATIONS

- Treatment of initial and recurrent HSV genital infections
- Suppression of frequently recurring genital HSV infections
- Treatment of herpes zoster

DOSING REGIMENS

INITIATION OF THERAPY

As with its parent compound, acyclovir, valacyclovir therapy should be initiated as soon as possible after the appearance of an HSV or VZV rash.

RISKS AND PRECAUTIONS

In addition to the effects noted with acyclovir, thrombotic thrombocytopenic purpura/hemolytic uremic syndrome (TTP/HUS) has been observed in patients with advanced HIV disease and in transplant recipients receiving dosages of 8 g/day.

COMPLICATIONS

Adverse effects are those that would be expected with acyclovir. However, severe and even fatal cases of TTP/HUS have been reported in patients with acquired immunodeficiency syndrome (AIDS) and transplant recipients taking high dosages of valacyclovir. The basis for these events and their causal relationship to valacyclovir have not been established. TTP/HUS has not been reported in patients taking conventional dosages (up to 3 g/day) of valacyclovir.

TABLE 232-2: Dosing Regimens for Valacyclovir

CONDITION	DOSAGE	ROUTE	DURATION
Primary HSV infection, immunocompetent host	1000 mg bid	Oral	10 days
Recurrent genital HSV infection, immunocompetent host	500 mg bid	Oral	3 days
Recurrent oral labial HSV infection, immunocompetent host	2000 mg bid	Oral	1 day
Oral labial HSV infection, immunocompromised host	500 mg bid	Oral	5–10 days
Chronic suppression of HSV infection	1000 mg qd	Oral	Undefined/indefinite
Varicella: adolescent/adult	1000 mg tid	Oral	5 days
Herpes zoster: normal host or not severe disease in immunocompromised host	1000 mg tid	Oral	7 days

HSV = herpes simplex virus.

Famciclovir and Penciclovir

MECHANISM OF ACTION

Famciclovir (9-[4-hydroxy-3-hydroxymethylbut-1-yl] guanine) is a prodrug of penciclovir (9-[4-hydroxy-3-hydroxymethylbut-1-yl] guanine). Penciclovir is phosphorylated to penciclovir triphosphate. The initial phosphorylation of penciclovir to penciclovir monophosphate is mediated by HSV- or VZV-induced thymidine kinases, similar to the activation of acyclovir. Subsequent phosphorylation to the diphosphate and triphosphate forms is carried out by cellular kinases. Penciclovir triphosphate inhibits viral DNA polymerases and interferes with extension of the nascent viral DNA chain. Due to the presence of a hydroxyl group on the acyclic side chain of penciclovir, limited DNA chain extension may still occur.

PHARMACOKINETICS

Famciclovir is available as an oral formulation and is converted to penciclovir by deacetylation and oxidation in the liver and intestine. The bioavailability of famciclovir is 77%. After oral administration of 500 mg famciclovir, a peak plasma concentration of penciclovir (3.3 µg/mL) is reached within 1 hour.